

Gargi Mitra

Networks & Systems Security Researcher | AI/ML Security, Vulnerability Research, Telemetry Analysis

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Professional Summary

I am a security researcher with a PhD in network security and privacy, and Postdoctoral experience in the security of AI-enabled safety-critical systems and the AI-enabled security assessment of heterogeneous networked systems. I specialize in correlating design decisions and security mechanisms across layers of the technology stack and incorporate interdisciplinary insights to uncover novel, high-impact vulnerabilities that conventional analysis often misses. Specifically, my work pivots on a prevalent practical concern: what new security risks emerge when a new technology is integrated into infrastructure that was not designed for it? I pursue these risks end to end, from threat modeling and testbed development to working proof-of-concept exploit against a real system and the mitigations that follow.

This approach has led me to uncover how application and protocol design choices influence information leakage through network-layer metadata, such as packet sizes, timing, and ordering, even when the traffic is encrypted. Building on this, I developed the first encrypted-traffic-analysis techniques for mass surveillance that adapt to modern web protocols (Snoopy, HTTP/2 fingerprinting). I also recovered viewers' interaction data from Netflix's interactive streaming using encrypted traffic metadata alone (White Mirror). In industrial control systems, I showed that the deterministic, periodic communication of industrial protocols lets an attacker identify process-critical packets within encrypted ICS/OT traffic and drop them to disrupt the physical process while evading state-of-the-art detectors (ICS-Sniper, the first targeted blackhole attack on encrypted ICS/OT traffic). This work draws on in-depth knowledge of TCP/IP internals, encryption techniques, VPN technologies, SCADA/Modbus/CIP, state-of-the-art attack detectors, and safety and security constraints.

Beyond network traffic, I study multi-vendor, AI-enabled networked systems, including safety-critical systems, where AI model designers cannot anticipate which third-party devices will later be integrated or how those integrations will change the attack surface of the system. To address this, I built LLM- and RAG-driven tooling that combines System-Theoretic Process Analysis (STPA) with vulnerability knowledge bases such as CVE and MITRE ATT&CK and ATLAS to derive attack paths directly from a system's design artifacts (SAM, MedAIScout). Across all of this work, components that are secure in isolation are shown to become vulnerable once they are combined. Agentic AI deployed in production systems is the most recent and highest-stakes instance of this pattern, and it is where I intend to focus next.

I regularly translate research findings into actionable insights for technical and non-technical audiences. My work has been published at top-tier systems and security venues (NDSS, IEEE TDSC, IEEE/ACM CHASE, EuroSys, IEEE/IFIP DSN), covered by WIRED, and recognized with awards. I report findings to industry executives (Medtronic, GE, Medcrypt) and regulators (the FDA), and I deliver quarterly reports to the National Cybersecurity Consortium, the Department of National Defence, and the National Research Council of Canada. I also coordinate a five-member team across concurrent projects and review technical work for ACM CCS and ASIACCS.

RELEVANT COMPETENCIES AT A GLANCE

AI/ML and Agentic System Security (≈ 4 years).

- Systematized the identification of inference-time security risks in AI/ML-based systems arising from integration with legacy infrastructure; defined the threat models and built the automated tooling for risk identification
- *SAM, SAMD*: Automated LLM-assisted identification of attack paths at design time in AI/ML-enabled systems, using STPA
- *MedAIScout*: LLM+RAG tool for automated retrieval of known AI/ML vulnerabilities from system design (second prize, ICICS Ideas with Impact, UBC 2024)
- *WireHound (ongoing)*: AI-enabled automated vulnerability identification from heterogeneous network traffic, with minimal system information
- *ROAST*: Risk-aware outlier exposure for adversarial selective training of anomaly detectors against evasion attacks (under review); related risk-profiling defence for DNNs presented at DSML'25

- Organized a summer workshop on AI security at UBC (2025); participated in a workshop on AI safety and security in Microsoft Redmond (June 2026)

Vulnerability Discovery and Attack Development (10+ years).

Designing, implementing, and validating novel attacks that expose emerging vulnerability classes in safety- and privacy-critical systems, across web, industrial, medical device, and IoT telemetry

- *ICS-Sniper*: First targeted blackhole attack on encrypted traffic of industrial OT networks using cloud-hosted SCADA; built the hardware and software testbeds and demonstrated resilience to state-of-the-art detectors
- *Snoopy*: Encrypted traffic analysis (ETA)-based mass-surveillance framework; privacy attacks on HTTP/2 multiplexing, showing that protocol-level multiplexing does not provide the privacy guarantee it was assumed to
- *White Mirror*: First side-channel attack on Netflix's interactive streaming traffic, recovering user interaction data from encrypted traffic metadata
- IDS/IPS evasion analysis; custom pcap analysis, tshark/Wireshark, Zeek, live flow processing for near-real-time attack detection

Project leadership, cross-team coordination, and program delivery.

- Lead researcher for 2.5+ years in a 5-member team at UBC working on the security of cloud-connected ICS OT; co-authored a grant proposal awarded CAD 500K by the National Cybersecurity Consortium of Canada
- Owned research roadmap, deadlines, and deliverables across simultaneous projects: problem scoping, equipment procurement, testbed setup, experiment planning, authoring papers and progress reports, delivering talks
- Organized Birds-of-a-Feather sessions bridging industrial and academic CPS security research at USENIX Security 2023 and 2025
- Mentored 4 graduate students (1 PhD, 2 MSc, 1 collaborator) and 10+ undergraduates and interns
- Experienced working both on location and in hybrid environments, with minimal supervision

Telemetry, tooling, and automation. Every attack above was mounted on, or had to evade, live network telemetry — packet size, timing, and burst structure over TLS; I build the data collection and analysis tools myself: Python and Bash pipelines over Zeek, tshark/Wireshark, NetfilterQueue and custom pcap tooling for near-real-time flow processing and inline interception; ICS and web testbeds with automated experiment orchestration; and LLM/RAG tooling that replaces manual literature and vulnerability review. Each attack is benchmarked against state-of-the-art IDS/IPS and anomaly detectors. Deriving early-warning indicators for AI-driven attacker behaviour is the same problem on different telemetry.

Technical reporting, review, and editorial judgment.

- 10+ peer-reviewed publications at top security and systems venues (NDSS, IEEE TDSC, Springer Nature, DSN, EuroSys, INFOCOM, workshops at Usenix Security, ACM CCS), interdisciplinary venues (IEEE/ACM CHASE), and AI workshops (NeurIPS)
- Presentations and written reports for industry executives (Medtronic, GE, Medcrypt) and regulators (the FDA); quarterly reporting to funding agencies (Department of National Defence of Canada, National Research Council of Canada, National Cybersecurity Consortium of Canada)
- Invited talks on emerging ICS OT security threats at Idaho National Laboratory and ETS Montreal
- Media coverage and interview in [Wired](#), [Financial Express](#), [Gadgets 360](#)
- Information dissemination to non-specialist audiences via [blog writing](#) (awarded by IBM India and the Govt. of India), [webinars](#)
- ACM CCS 2024 Distinguished Reviewer Award (top 10% of 527 PC members); PC member: CCS 2024/25/26, ASIACCS 2026; admission reviewer for African Computer Vision Summer School 2025/26

Work Experience

Postdoc-

Postdoctoral Research Fellow

Jan 2023–Aug 2026

The University of British Columbia (UBC), Vancouver, Canada

- Advisors: Karthik Pattabiraman (Dept. of ECE), Aastha Mehta (Dept. of CS)
- Research Area: Identifying emerging security and privacy vulnerabilities in modern safety-critical systems
- Research highlights:

1. ICS-Sniper: A Targeted Blackhole Attack in Cloud-Connected Industrial Control Systems

Lead Researcher and Mentor

- Accepted for presentation at NDSS 2027
- Designed and implemented the first targeted blackhole attack on encrypted ICS/OT network traffic, and suggested countermeasures at different levels of the communication stack
- Built the state-of-the-art hardware- and software-based testbeds; authored the papers
- Led the 5-member multi-year project; mentored teammates
- Co-authored the grant proposal with the PIs, awarded CAD 500K by the National Cybersecurity Consortium (2024); reported progress to the NCC quarterly
- Delivered talks at Idaho National Laboratory and ETS Montreal; engaged with industry and academic security practitioners

2. Security of AI/ML-Enabled Medical Devices

Researcher and Mentor

- Systematized the identification of inference-time security risks in AI/ML-based medical devices, emerging from the integration of AI/ML techniques with legacy infrastructure
- Defined threat models and built LLM+RAG-enabled automated tools for identifying inference-time security risks from system design, informed by ML attack literature, MITRE ATLAS framework, and CVE databases
- Collaborated with and mentored PhD, MASc, undergraduate, and Capstone students from UBC, University of Virginia, and IIT Madras
- Wrote reports and delivered presentations for academic venues, industry executives (Medtronic, GE, Medcrypt), and the FDA
- Published at IEEE/IFIP DSN and NeurIPS workshops, IEEE/ACM CHASE, and Springer Nature HealthSec

3. IoT Privacy & Information Flow Control

Collaborating Researcher

- Contributed privacy research ideas to and co-authored *Turnstile*, a hybrid information flow control framework for GDPR-compliant IoT applications
- Co-authored *ImmunoPlane*, middleware providing adaptivity to distributed IoT applications

Programmer Analyst

2013–2015

Cognizant Technology Solutions India Pvt.Ltd. (Domain: Data warehousing)

Academic Background

PhD

PhD and MS under Dual Degree Program (Computer Science and Engineering)

2015–2022

Indian Institute of Technology Madras (IIT Madras), Chennai, India

- Advisors: Kamakoti Veezhinathan, Nitin Chandrachoodan
- Dissertation Title: Encrypted Traffic Analysis-Based Cyber Surveillance and Possible Mitigation Techniques
- Research highlights:
 1. Snoopy: A Webpage Fingerprinting Framework for Mass Surveillance.
 2. First Encrypted Traffic Analysis-Based Attack on HTTP/2 Traffic.
 3. White Mirror: The First Side-Channel Attack on Netflix's Interactive Streaming Traffic.
 4. Security and QoS measurement research in untrusted multi-vendor service function chains
 5. Additional collaborations: Kernel implementation of SCION, a secure Internet communication protocol, in collaboration with Prof. Adrian Perrig, ETH Zurich

Mitacs-SICI Indo-Canadian Research Internship

Sep 2021–Dec 2021

UBC + IIT Madras

- Research Internship Topic: Encrypted Traffic Analysis-based attacks on Internet-connected ICS

Bachelor in Technology (Computer Science and Engineering)

2009–2013

Top Awards and Recognitions

- 2024** • *Ideas with Impact Competition, **second prize***, issued by UBC, for presenting MedAIScout, a tool for automating cybersecurity assessment of ML-enabled medical devices. It is an annual competition that invites **graduate students and postdocs across all departments of UBC** to showcase their research from the perspective of commercial potential.
- 2021** • *Mitacs Globalink Research Award*, sponsored by Mitacs and Shastri Indo-Canadian Institute, for collaborative research between India and Canada, and **open to all Doctoral students of India**.
- 2020** • *Applied Research Competition, **first prize***, at Cyber Security Awareness Week (CSAW), organized by NYU and IIT Kanpur, **open to all PhD students across India working in the domain of cybersecurity**.
- 2019** • *AWSAR top 100 Award*, awarded to the **top 100 out of 4993** contestants in a national scientific story-writing competition organized by the Department of Science & Technology, Govt. of India.
- 2017** • *Embedded Security Challenge Winner, **first prize*** at Embedded Security Challenge, Cyber Security Awareness Week, organized by NYU and IIT Kanpur, for presenting 'Eradicator: An Integrated Approach for Defense against Cyber Attacks in PLC based ICS'. The competition was **open to students of all Indian universities**.

Peer-Reviewed Publications

Security of AI/ML-Based Systems

1. **'Red Teaming GenAI' workshop (co-located with NEURIPS'24)** *MedAIScout: Automated Retrieval of Known Machine Learning Vulnerabilities in Medical Applications*, [Paper]
2. **Springer Nature HealthSec 2024 Journal** *Systems-Theoretic and Data-Driven Security Analysis in ML-enabled Medical Devices*, (Invited) [Paper] — won 2nd prize at UBC's Ideas with Impact competition for commercial potential.
3. **DSML'25 workshop (co-located with DSN'25)** *Learning from the Good Ones: Risk Profiling-Based Defenses Against Evasion Attacks on DNNs*, [Acceptance rate: 64%] [Paper]
4. **HealthSec'24 workshop (co-located with ACM CCS'24)** *SAM: Foreseeing Inference-Time False Data Injection Attacks on ML-enabled Medical Devices*, [Acceptance rate: 57%] [Paper]
5. **IEEE/ACM CHASE'24** *Systematically Assessing the Security Risks of AI/ML-enabled Connected Healthcare Systems*, [Acceptance rate: 28.4%] [Paper]
6. **[Under review]** *ROAST: Risk-aware Outlier-exposure for Adversarial Selective Training of Anomaly Detectors Against Evasion Attacks*, [Paper]
7. **[In Pipeline]** *SAMD: A Tool for Identifying False Data Injection Scenarios in AI/ML-enabled Medical Devices*, [Paper]

Web Security & Encrypted Network Traffic Analysis

1. **NDSS'27** *ICS-Sniper: A Targeted Blackhole Attack on Encrypted ICS Traffic*, [Paper]; also presented at RICSS'24 workshop (co-located with ACM CCS'24), USENIX '25 and Idaho National Laboratory. [Extended version under review]
2. **IEEE TDSC'22** *Snoopy: A Webpage Fingerprinting Framework with Finite Query Model for Mass-Surveillance*, [Acceptance rate: 10-12%, Impact factor: 7.5] [Paper]
3. **IEEE/IFIP DSN'20** *Depending on HTTP/2 for Privacy? Good Luck!*, [Acceptance rate: 16.5%] [Paper]
4. **ACM SIGCOMM'19 Extended Abstract** *White Mirror: Leaking Sensitive Information from Interactive Netflix Movies using Encrypted Traffic Analysis*, [Acceptance rate: 55%] [Paper] – Sensitive information leakage from Netflix via encrypted traffic — covered by (a) **Wired**, (b) **Financial Express**, (c) **Gadgets 360**; informs data privacy risk advisory.
5. **IEEE INFOCOM'19** *Towards Measuring Quality of Service in Untrusted Multi-Vendor Service Function Chains: Balancing Security and Resource Consumption*, [Acceptance rate: 19.7%] [Paper] – Security/resource trade-offs in untrusted network service function chains.
6. **IEEE MCOMSTD'18** *ApproxBC: Blockchain Design Alternatives for Approximation-Tolerant Resource-Constrained Applications*, [Paper]

Security of IoT Applications

1. **EuroSys'26** *Turnstile: Hybrid Information Flow Control Framework for Managing Privacy in Internet-of-Things Applications*, [Acceptance rate: 17%] [Paper]

2. **IoTDF'24** *ImmunoPlane: Middleware for Providing Adaptivity to Distributed Internet-of-Things Applications*, [Acceptance rate: 36.7%] [\[Paper\]](#)